Product Requirements Document

VoteReady: a voter information web app powered by the Google Civic Information API
Author: Oriana | ISM6427 AI Business Innovation | Version 1.0 | October 27, 2026

1. Overview

VoteReady helps people get ready to vote. A user enters their home address and gets everything they need for the next election in one place: where to vote, what is on their ballot, plain language explanations of ballot measures, a personal voting plan, reminders, and links to their election office. The app runs entirely on Netlify with no database.

Important scope note: VoteReady does not let people cast real ballots. U.S. public elections do not allow online voting through third party apps, so the app prepares and guides voters, and sends them to official channels to register, request a mail ballot, and vote.

2. Problem and goals

Voting information is spread across state sites, county sites, and mailers. First time and busy voters often do not know where to vote, what is on the ballot, or what a ballot measure actually does. Goals:

- Get a voter from address to a complete voting plan in under 2 minutes.
- Explain every ballot measure in plain, neutral language.
- Stay free to run: static hosting, serverless functions, no database, no user accounts.
- Stay nonpartisan: show official data only, no endorsements or rankings.

3. Target users

Persona	Need
First time voter (18 to 24, college student)	Simple steps: am I registered, where do I go, what is on my ballot
Busy professional	Fast answer and a reminder so they do not forget
Community leader (church, campus, nonprofit)	A page to share with their group so members can look up their own info
Spanish speaking voter in South Florida	The same information in Spanish (phase 3)

4. Features (the 8 core features)

#	Feature	What it does	Data source
1	Polling place finder	Address search shows polling places, early voting sites, and drop boxes on a map with hours and directions links.	/voterinfo: pollingLocations, earlyVoteSites, dropOffLocations
2	Personalized sample ballot	Shows every contest on the user's ballot: offices, candidates (name, party, website, social links), and measures. User can mark choices privately to bring to the polls.	/voterinfo: contests

#	Feature	What it does	Data source
3	Ballot measure explainer (AI)	One tap turns official measure text into a neutral plain language summary: what a Yes vote does, what a No vote does, plus the official pro and con statements.	/voterinfo: referendumText, pro and con statements + Claude API
4	Voting plan builder	User picks in person, early, or by mail and gets a step checklist with deadlines and the right official links. Printable and saved on the device.	/voterinfo: state.electionAdministrationBody, mailOnly
5	Election countdown and reminders	Lists upcoming elections with a countdown. Reminder button downloads a calendar file (.ics) for election day and early voting.	/elections
6	Election office and registration hub	Shows the user's state and local election office: register or check registration, absentee info, ballot info, hours, contact details.	/voterinfo: state, electionAdministrationBody, electionOfficials
7	District explorer	Shows which political districts the address belongs to (congressional, state house, county, city, school) and lets users search divisions by name.	/divisions + OCD IDs from /voterinfo
8	Community hub and embed	Shareable link and an embeddable widget a church, campus, or group can put on its site. Each visitor looks up their own address.	All of the above, configured by URL parameters

5. User stories

- As a voter, I enter my address and see my polling place on a map so I know where to go.
- As a voter, I see my sample ballot so I can research candidates before election day.
- As a voter, I tap a ballot measure and read a short, neutral explanation of what Yes and No mean.
- As a voter, I choose how I will vote and get a checklist with my deadlines.
- As a voter, I add election day to my calendar in one tap.
- As a voter, I find my election office's website and phone number to check my registration.
- As a voter, I see which districts I live in.
- As a community leader, I copy a link or embed code and share it with my group.

6. Technical architecture (Netlify, no database)

Layer	Choice	Notes
Frontend	Static site (HTML, CSS, JavaScript; React with Vite optional)	Deployed from GitHub to Netlify on every push to main.
API proxy	Netlify Functions: /api/elections, /api/voterinfo, /api/divisions	Keeps the Civic API key secret on the server. Adds cache headers.
AI explainer	Netlify Function /api/explain calling the Claude API	Prompt requires neutral wording, Yes and No effects, and no recommendation.
Maps	Leaflet with OpenStreetMap tiles	Free, no extra key. Coordinates come from the API response.

Layer	Choice	Notes
State	Browser localStorage and URL parameters	Saved address, ballot picks, and plan live only on the user's device. Clear my data button.
Caching	Netlify CDN with Cache-Control headers	Voter info and contests up to 24 hours, elections list 1 hour, divisions up to 30 days, per Google's data guidelines.
Reminders	Client side .ics file download	No email or SMS, so no database or contact list is needed.
Secrets	Netlify environment variables	CIVIC_API_KEY and ANTHROPIC_API_KEY. Never shipped to the browser.

7. Non functional requirements

- **Privacy:** addresses are never stored or logged on the server. No accounts, no tracking cookies.
- **Neutrality:** candidates shown in the order the API returns them; no endorsements; AI summaries labeled as AI generated with the official text one tap away.
- **Accessibility:** WCAG 2.1 AA, keyboard friendly, screen reader labels, high contrast.
- **Performance:** mobile first, first screen loads in under 2 seconds on 4G.
- **Accuracy:** every screen reminds users to confirm with their local election office, and links to it.
- **Error handling:** friendly messages for each API status (address not found, no data yet, election over).

8. API constraints and how the app handles them

Constraint	Plan
Only elections covered by the Voting Information Project	Show a clear message and the election office hub when no data exists.
Elections disappear after election day	Off season mode: countdown to the next listed election plus the office and district features.
Representatives API was shut down in April 2025	Out of scope for v1. Phase 3 option: OCD IDs plus OpenStates or BallotReady for a who represents me page.
Quota increases take 1 to 2 months	Request extra quota by August before November elections; CDN caching lowers calls.
Testing outside election season	Use election ID 2000, Google's permanent test election, in a demo mode.

9. Release plan

Phase	Scope	Done when
1: MVP	Features 1, 2, 6, address search, Netlify Functions proxy, demo mode	A real address returns a polling place, ballot, and office info on the live Netlify URL.
2: Engage	Features 3, 4, 5 (AI explainer, plan builder, reminders)	A user can finish a plan and add election day to their calendar.

Phase	Scope	Done when
3: Share	Features 7, 8, Spanish language, optional representatives page	A group can embed the widget on its own site.

10. Success metrics

- Percent of address searches that return a polling place (target 90% during covered elections).
- Percent of users who finish a voting plan (target 40%).
- Calendar reminder downloads per 100 visitors (target 25).
- Measure explainer opens per ballot viewed.
- Measured without a database using Netlify Analytics (page level, no personal data).

11. Risks

- **AI error in a measure summary:** strict neutral prompt, show the official text right below, label as AI, cache one summary per measure.
- **Data gaps or late changes from election offices:** verify message and office link on every screen.
- **API cost or quota spikes near election day:** CDN caching and a rate limit in each function.
- **Perceived bias:** official data only, consistent design for every candidate and party.

12. Out of scope (v1)

- Casting or submitting real ballots.
- User accounts, a database, or email and SMS reminders.
- Election results and polling data.
- Candidate endorsements or recommendations.